

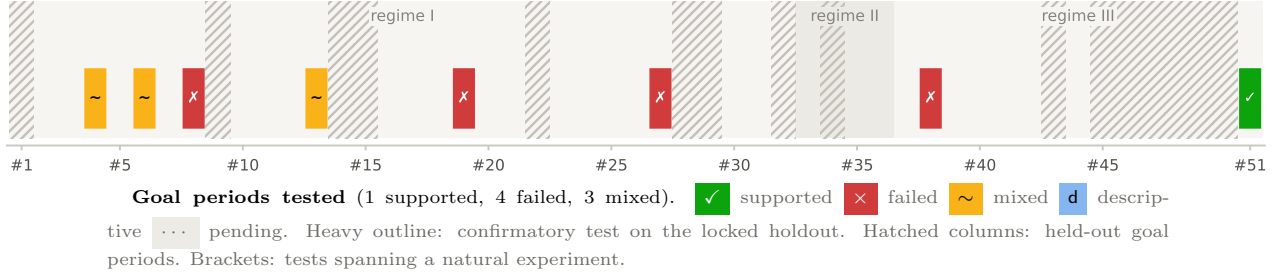
H120: Is a goal period a stationary NESS? Within-period drift of J and EP (#38, #51 main)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 3/5 Usefulness 2.5/5

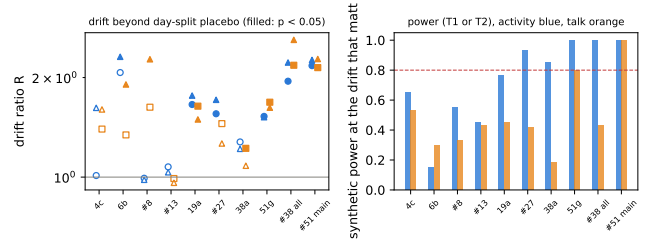
Bottom line. Inside long goal periods the kinetic Ising parameters of activity and talk drift beyond day-split placebo noise (#38: R 1.95 and 2.18, Holm ≤ 0.008 ; #51: 2.17 and 2.14), with no step at NE17, while an 8-day unit shows no drift; entropy-production drift is not identifiable.

level: descriptive · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Every hypothesis in this project fits its model inside one goal period and treats the fit as one point on a phase diagram. That assumes the period is a stationary nonequilibrium steady state. HH360 tests the assumption on the longest periods: the 17-day charity drive (#38), predicted stationary, and the #51 main body (45 days), predicted to drift. H120 fits a kinetic Ising model to 1-min activity and talk spins of the core agents and asks whether the first and second halves (or a linear trend) differ more than random day halves do. **#38 is not stationary, and the HH's kill fires.** Its halves differ about twice as much as random day halves in both channels. The drift is gradual, not a step at NE17. The 8-day unit 38a shows no drift.



Observables. Left: drift ratio R for the contiguous split (circles, squares) and the linear trend (triangles) per window; activity blue, talk orange; filled = permutation $p < 0.05$. $R \approx 1$ is a stationary window. #38, #51 main, 51g, #27 and 19a drift; 38a, #13, 4c do not. Right: synthetic power at the drift that matters (a re-randomization of J at its own scale over the window); the dashed line marks 0.8. Regime-I units with 4–6 agents are underpowered.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 02 (kinetic Ising, logistic Glauber updates on a 1-min grid inside the DQ8 all-present window). Core agent i updates with field h_i , couplings J_{ij} (incl. persistence J_{ii}), weekday fields $\phi_{i,w}$ and a session-length field ψ_i , ridge 1 on J in every window. Drift test: an efficient score statistic W for $\theta_i(d) = \theta_i + \tau_d \delta_i$ on (h_i, J_i) , from per-day gradients g_d and Hessians H_d at the pooled fit, calibrated by permuting days. Entropy production: the corrected held-out Newton bound on antisymmetric lagged pair and single-agent observables.

For anyone fitting models to a swarm's logs, including this project: a goal period is not one steady state. Kinetic Ising parameters move by about 0.2–0.4 placebo SD per day, so a 17-day period fit averages over a walking state. Two rules follow. Fit in windows of about one week (the 8-day unit 38a passes; the 13-day unit 51g does not). Report long-period parameters as trajectories or weekly points, not as one constant. Step changes do not explain the drift: NE17 is an ordinary boundary.

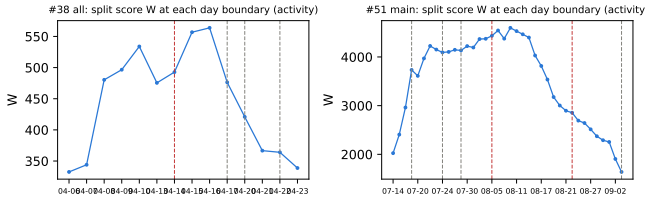
$$\text{logit } P(s'_i = 1 \mid \mathbf{s}) = h_i + \sum_j J_{ij} s_j + \phi_{i,w(d)} + \psi_i L_d$$

$$U_i = \sum_d \tau_d P g_{i,d}, \quad W = \sum_i U_i^\top V_i^{-1} U_i, \quad R = W_{\text{obs}} / (W)_{\text{random day splits}}$$

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RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: size, power, step rank	size ≤ 0.12 per test; activity power 1.00 (#38, #51), 0.85 (38a); EP power ≤ 0.30 ; step top-10% 100%	passed (EP unpow-ered)
P1 #38 stationary (HH)	R 1.95 / 2.18; trend 2.22 / 2.60; Holm ≤ 0.008 ; robust to weekday strata and day-1 drop	failed (kill)
P2 38a stationary	R 1.28 / 1.22; Holm ≥ 0.16 ; power 0.85	supported
P3 #51 main drifts	R 2.17 / 2.14, p 0.001	supported
P4 boundaries top 20%	NE17 0.36 / 0.21; #51 08-05 0.13 (activity)	failed / supported
P5 replication: no drift $\geq 6/8$	4/8 (3 underpowered); 8, 19a, 27, 51g drift	failed
P6 EP stable in #38, drifts in #51	no EP drift detected anywhere testable	inconclusive
HH360	#38 drifts; #51 drifts; no NE17 step	failed



Second figure. Split score W at every day boundary of #38 (left) and the #51 main body (right), activity channel. Dashed lines: step changes (red: the pre-registered ones). A step would make one boundary stand out; instead W rises smoothly toward the middle, the signature of a trend. Only #51's 08-05 boundary (#focus room, bookends stop) ranks in the top 13%.

GOAL PERIODS IN DETAIL

G04 ~ mixed
G06 ~ mixed
G08 × failed
G13 ~ mixed
G19 × failed
G27 × failed
G38 × failed
G51 ✓ supported

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The drift statistic includes the fields h_i as well as J ; post hoc, J alone also drifts in #38 and #51, and #38 drifts before its first roster join. Non-core agents (joiners) act as an unmodelled field. Talk is underpowered in most windows, and EP drift is unpowered everywhere; the 51g EP rejection rests on negative bounds and is not read as an arrow of time. First-order Markov dynamics on a 1-min grid is assumed. Regime-I units with 4-6 agents cannot tell stationarity from drift.

ROUND 2

Fit $J(t)$ with a random-walk prior and report a drift rate per day; estimate the longest stationary window per regime; regress the weekly drift on roster churn, operator volume and project churn; run the same test on H91's content modes. Confirmation on the #51 tail and unit 1a (frozen, not run).

- **Essence.** how fast a swarm's dynamical parameters walk inside a period, and what window a stationary fit can use.
- **H120-R1** Time-varying parameters: fit $J(t)$ with a random-walk prior (Kalman or GP over days), report the drift rate per day with an interval, and place periods on the phase diagram as trajectories.
- **H120-R2** Stationary window length: estimate the longest window whose split test passes, per regime (38a passes at 8 days; 51g fails at 13).
- **H120-R3** Drivers of the walk: regress the weekly one-step δJ on roster share outside the core, operator message volume and project churn (DQ4).
- **H120-R4** Content analog: the same score test on H91's content modes, to tie the two "about 1 SD a day" drifts.

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